



GENERATIVE AI

ASSIGNMENT NO 1

LLM CHATBOT USING GOOGLE GEMINI

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TECHNOLOGIES USED

Python: Programming language used to develop the chatbot.

Google Colab: Online environment used to write and execute the Python program.

Google AI Studio: Used for working with Google's Gemini API.

Gemini: The language model used to generate responses.

Google Generative AI SDK: Used to interact with the Gemini model through Python.

LangChain: Framework used for developing applications based on language models.

Gemini API: Provides access to the Gemini model.

PROGRAM / CODE

LLM Chatbot Program

```
# Install required libraries

!pip install -U langchain-google-genai

!pip install -U langchain


# Import libraries

import google.generativeai as genai

from google.colab import userdata


# Get API key securely from Google Colab

GOOGLE_API_KEY = userdata.get('Vaishnavi')

genai.configure(api_key=GOOGLE_API_KEY)


# Initialize Gemini model

gemini_model = genai.GenerativeModel('gemini-3.5-flash')


# Interactive LLM Chatbot

while True:

    user_question = input("You: ")

    if user_question.lower() == "exit":

        print("Chatbot ended.")

        break

    response = gemini_model.generate_content(user_question)

    print("Gemini:", response.text)
```

OUTPUT

```
[1] 6s !pip install -U langchain-google-genai
... Requirement already satisfied: langchain-google-genai in /usr/local/lib/python3.13/dist-packages (4.4.0)
Requirement already satisfied: filetype<2.0.0,>=1.2.0 in /usr/local/lib/python3.13/dist-packages (from langchain-google-genai) (1.2.0)
Requirement already satisfied: google-genai<3.0.0,>=2.20.0 in /usr/local/lib/python3.13/dist-packages (from langchain-google-genai) (2.22.0)
Requirement already satisfied: langchain-core<2.0.0,>=1.6.1 in /usr/local/lib/python3.13/dist-packages (from langchain-google-genai) (1.6.1)
Requirement already satisfied: pydantic<3.0.0,>=2.0.0 in /usr/local/lib/python3.13/dist-packages (from langchain-google-genai) (2.13.4)

!pip install -U langchain
... Requirement already satisfied: langchain in /usr/local/lib/python3.13/dist-packages (1.3.17)
Collecting langchain
  Downloading langchain-1.4.0-py3-none-any.whl.metadata (6.2 kB)
Requirement already satisfied: langchain-core<2.0.0,>=1.6.0 in /usr/local/lib/python3.13/dist-packages (from langchain) (1.6.1)
Requirement already satisfied: langgraph<1.3.0,>=1.2.11 in /usr/local/lib/python3.13/dist-packages (from langchain) (1.2.11)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.13/dist-packages (from langchain) (2.13.4)
Requirement already satisfied: httpx<1.0.0,>=0.23.0 in /usr/local/lib/python3.13/dist-packages (from langchain-core<2.0.0,>=1.6.0->langchain) (0.27.0)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.13/dist-packages (from langchain-core<2.0.0,>=1.6.0->langchain) (1.33.0)

[3] 2s import google.generativeai as genai

/usr/local/lib/python3.13/dist-packages/google/colab/_import_hooks/_hook_injector.py:55: FutureWarning:

All support for the `google.generativeai` package has ended. It will no longer be receiving updates or bug fixes. Please switch to the `google.genai` package as soon as possible. See README for more details:
https://github.com/google-gemini/deprecated-generative-ai-python/blob/main/README.md

  loader.exec_module(module)

[4] 0s from google.colab import userdata

[5] 1s GOOGLE_API_KEY = userdata.get('Vaishnavi')

[6] 0s genai.configure(api_key=GOOGLE_API_KEY)

[7] 0s gemini_model = genai.GenerativeModel('gemini-3.5-flash')

[8] 0s prompt_text = "What is Generative AI?"

[9] 9s response = gemini_model.generate_content(prompt_text)

[10] 0s print(response.text)
... **Generative Artificial Intelligence (Generative AI)** is a branch of artificial intelligence capable of generating new content—such as text, images, music, audio, video, or
Unlike traditional AI, which is designed to **analyze data or make decisions** (like spam filters or product recommendations), Generative AI is designed to **create something
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### How Does Generative AI Work? (In Simple Terms)

Generative AI works using **machine learning models**, specifically deep learning, trained on massive datasets.

1. **Training:** The AI is fed billions of examples of existing content (e.g., books, articles, photos, songs, or code).
2. **Learning Patterns:** The AI analyzes this data to learn the underlying patterns, structures, and relationships. For example, it learns that the word "bacon" often follow
3. **Prompting and Predicting:** When a user inputs a prompt (e.g., "Write a poem about a robot"), the AI doesn't copy-paste existing work. Instead, it uses probability to
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### Common Types and Examples of Generative AI

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Generative AI is categorized by the type of content it produces:

* **Text Generation:** Models like **ChatGPT** (OpenAI), **Claude** (Anthropic), and **Gemini** (Google) can write essays, answer questions, summarize long documents, translate, and more.
* **Image Generation:** Tools like **Midjourney**, **DALL-E 3**, and **Stable Diffusion** can create highly realistic photos, digital art, or 3D models from a simple text description.
* **Code Generation:** Tools like **GitHub Copilot** can write, debug, and explain programming code in dozens of computer languages.
* **Audio & Music Generation:** Platforms like **Suno** or **ElevenLabs** can generate complete songs with vocals, or create highly realistic text-to-speech voiceovers.
* **Video Generation:** Tools like OpenAI's **Sora** or **Runway** can generate realistic video clips from text prompts.

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### Real-World Applications

Generative AI is rapidly transforming various industries:

* **Business & Marketing:** Writing copy, brainstorming ideas, and creating personalized advertisements.
* **Customer Service:** Powering advanced chatbots that can understand and solve complex customer issues.
* **Healthcare:** Helping researchers design new proteins and accelerate drug discovery.
* **Entertainment:** Assisting game developers in creating virtual worlds and helping writers story-board films.
* **Education:** Serving as personalized tutors that can explain complex topics in different ways tailored to the student.

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### The Challenges and Risks of Generative AI

While powerful, Generative AI has several limitations and ethical concerns:

* **Hallucinations:** Sometimes, these models confidently present false or fabricated information as fact.
* **Copyright and Ethics:** Because AI is trained on data created by humans (artists, writers, programmers), there are major ongoing debates and lawsuits regarding intellectual property.
* **Bias and Misinformation:** If the training data contains biases or inaccuracies, the AI will replicate them. It can also be used to create convincing "deepfakes" (fake audio or video).
* **Job Displacement:** There are concerns that Generative AI will replace human workers in creative and white-collar fields, such as writing, graphic design, and entry-level programming.

```

```

[11] 30s ▶ user_question = input("You: ")

response = gemini_model.generate_content(user_question)

print("Gemini:", response.text)

...
You: What is Deep Learning
Gemini: **Deep Learning** is a subfield of Artificial Intelligence (AI) and Machine Learning (ML) that is inspired by the structure and function of the human brain.

At its core, deep learning uses multi-layered artificial neural networks to learn from vast amounts of data. While traditional machine learning requires humans to manually label data, deep learning can automatically learn features from raw data.

Here is a breakdown of how it works, why it matters, and where it is used.

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### 1. Where Deep Learning Fits (The AI Hierarchy)
To understand deep learning, it helps to see how it fits into the broader picture of computer science:
* **Artificial Intelligence (AI):** The broad concept of enabling machines to mimic human intelligence.
* **Machine Learning (ML):** A subset of AI where computers learn from data without being explicitly programmed.
* **Deep Learning (DL):** A specialized subset of ML that uses "deep" neural networks to solve highly complex problems.

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### 2. How Deep Learning Works: Neural Networks
Deep learning gets its name from the "deep" layers of **Artificial Neural Networks (ANNs)**. These networks are modeled loosely after the biological neurons in the human brain.

A typical deep neural network consists of three main parts:
1. **The Input Layer:** Receives the raw data (e.g., an image of a handwritten number).
2. **The Hidden Layers:** This is where the "deep" part comes in. A network might have dozens or hundreds of these layers. Each layer extracts progressively higher-level features.
   * Example: If the network is analyzing a face, the first hidden layer might detect simple lines and edges. The next layer detects shapes (eyes, noses). The final hidden layer might detect facial features.
3. **The Output Layer:** Gives the final prediction or decision (e.g., "This is the number 7" or "This is a cat").

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```

[12] 1m ▶ while True:
    user_question = input("You: ")

    if user_question.lower() == "exit":
        print("Chatbot ended.")
        break

    response = gemini_model.generate_content(user_question)
    print("Gemini:", response.text)

...
You: what is AI
Gemini: At its simplest, **Artificial Intelligence (AI)** is the ability of a computer or a robot to perform tasks that usually require human intelligence and discernment.

While humans are born with natural intelligence, machines are programmed with artificial intelligence to mimic how humans **learn, reason, solve problems, and make decisions**.

Here is a breakdown of what AI is, how it works, and how it is shaping our world.

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### 1. How is AI different from normal computers?
* **Traditional Computers:** Follow strict, pre-written rules. (e.g., "If the user clicks this button, open that file.")
* **AI Systems:** Learn from experience and data. If you show an AI thousands of photos of cats, it will eventually figure out what a cat looks like on its own, without a human telling it.

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### 2. The Three Levels of AI
Scientists generally divide AI into three categories based on its capabilities:

1. **Artificial Narrow Intelligence (ANI) or "Weak AI":**
   * This AI is designed to do **one specific task** very well (e.g., play chess, recommend a movie, or drive a car).
   * Examples: Siri, Alexa, recommendation algorithms, self-driving cars.

```

* **Privacy:** AI requires massive amounts of personal data to work effectively, raising surveillance concerns.

Summary

AI is not a "living mind" or a sci-fi robot; it is a **powerful tool** that excels at finding patterns in data and using those patterns to make predictions, solve problems, You: what is machine learning

Gemini: At its simplest, **Machine Learning (ML)** is a branch of artificial intelligence (AI) that allows computers to learn and make decisions **without** being explicitly Instead of a human writing code with specific rules for every scenario, machine learning uses **algorithms** to analyze massive amounts of data, find patterns, and make pre

The Best Way to Understand It: Traditional vs. ML

To understand machine learning, it helps to compare it to traditional computer programming.

* **Traditional Programming:** You give the computer **Data** and **Rules** (the code), and the computer gives you the **Answers**.

* **Example:** If you want to block spam emails, you write a rule: "If the email contains the word 'viagra' or 'free money,' move it to the spam folder."

* **Machine Learning:** You give the computer **Data** and the **Answers**, and the computer figures out the **Rules** on its own.

* **Example:** You feed the computer 10,000 emails labeled "Spam" and 10,000 labeled "Not Spam." The ML algorithm analyzes them and detects subtle patterns (words, send

The Three Main Types of Machine Learning

Machine learning is generally divided into three categories, depending on how the algorithm learns:

1. Supervised Learning (The "Teacher-Student" Method)

The computer is trained on a "labeled" dataset. This means you feed it data where you already know the correct answer.

Where Do You Encounter Machine Learning Daily?

You likely use machine learning dozens of times a day without realizing it:

* **Entertainment:** Netflix, YouTube, and Spotify algorithms predicting what you want to watch or listen to next.

* **Smartphones:** Facial recognition (FaceID), predictive text, and voice assistants (Siri, Alexa).

* **Finance:** Banks using ML to instantly approve loans or flag suspicious credit card activity.

* **Transportation:** Uber calculating your ride fare and arrival time, or Google Maps predicting traffic.

AI vs. Machine Learning vs. Deep Learning

These terms are often used interchangeably, but they are actually nested inside one another:

1. **Artificial Intelligence (AI):** The broad concept of machines acting "smartly" (includes simple rules-based systems).

2. **Machine Learning (ML):** A subset of AI where machines learn from data rather than rules.

3. **Deep Learning (DL):** A advanced subset of ML that uses "neural networks" (mimicking the human brain) to solve highly complex problems, like translating languages or

You: exit

Chatbot ended.

PROGRAM DESCRIPTION

This program uses Python in Google Colab to create an AI-powered chatbot using the Gemini language model. The program takes a question from the user and sends it to the Gemini model through the Gemini API. The generated response is then displayed to the user. The API key is securely accessed using Google Colab's user data storage.

Working of the program:

1. The required Python libraries are installed.
2. The Google Generative AI SDK is imported.
3. The Gemini API key is accessed securely from Google Colab.
4. The Gemini model is initialized.
5. The user enters a question.
6. The question is sent to the Gemini model.
7. Gemini generates a response and displays it.
8. The chatbot continues until the user enters 'exit'.

REFERENCES

1. Google AI Studio
2. Gemini API Documentation
3. Google Gen AI Python SDK Documentation
4. LangChain Documentation

GITHUB LINK:

<https://github.com/VaishnaviSardar24/Generative-AI-Chatbot>